

Code-Layout, Readability and Reusability

Date	
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Code indentation is consistent across all files	Yes	Consistent indentation maintained throughout app.py, the Jupyter notebook, and HTML templates
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized with app.py, crop_model.pkl, scaler.pkl, requirements.txt, data/, templates/, and OptiCrop.ipynb
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	temperature, rainfall, humidity, ph, N, P, K
4	Function / Method Names	Functions are descriptively named	Yes	Functions used for model loading, input scaling, and crop prediction
5	Code Comments	Inline and block comments explain logic	Partial	Key ML and prediction functions are commented
6	Modular Design	Code is split into reusable functions/modules	Yes	Frontend templates, ML model, dataset preprocessing, and prediction logic are organized separately
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code is avoided by reusing functions and keeping the code clean
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and error handling implemented in the /predict route

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Medium/Low)
1	Crop prediction model	Python, scikit-learn	Crop prediction module	High
2	Data preprocessing & scaling module (StandardScaler)	Python, Pandas	Model training & prediction	High
3	Flask API	Python, Flask	Web application backend	High
4	HTML templates (Home / About / Find Your Crop / Result)	HTML, CSS	Farmer-facing web pages	Medium
5	Dataset loader	Python, Pandas	Training and testing process	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project is well organized with separate files for the ML model, dataset, application, and documentation.
Readability	5	Code uses meaningful variable names, proper indentation, and clear function names.
Reusability	5	Reusable modules like the crop prediction model, dataset loader, and preprocessing/scaling are used across the project.
Documentation / Comments	4	Basic comments and project documentation are provided, but more detailed inline documentation can be added.
Overall Score	5	The project follows good coding standards, modular design, and reusable components with a clean structure.